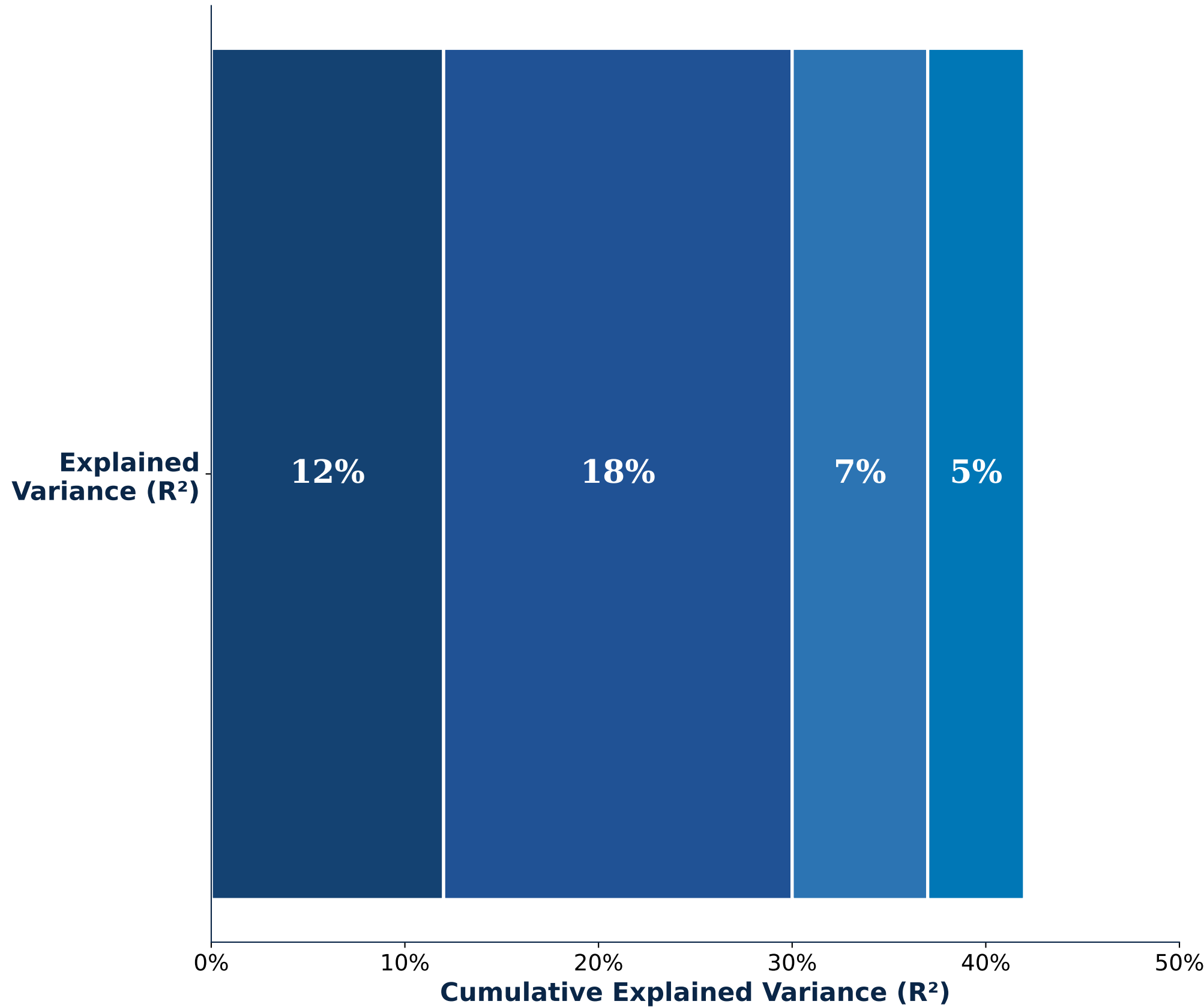


HIERARCHICAL MULTIVARIATE REGRESSION ANALYSIS

Epigenetic Age Acceleration Predictors (n = 3,847)

A. Variance Decomposition



B. Model Progression

Model 1	R² = 0.12	
Age + Sex		
↓		
Model 2	R² = 0.30	F = 287.4
+ Substance use duration	ΔR² = 0.18	p < 0.001
↓		
Model 3	R² = 0.37	F = 94.3
+ Physiological mediators (HOMA-IR, Cortisol/ACTH, Inflammation)	ΔR² = 0.07	p < 0.001
↓		
Model 4	R² = 0.42	F = 67.8
+ Psychological moderators (DERS, SCS-B)	ΔR² = 0.05	p < 0.001

KEY FINDINGS

Substance use (18%) + Physiological mediators (7%) + Psychological moderators (5%)

Together explain 30% of epigenetic aging variance beyond demographics

Demographics (Age, Sex): 12% | Substance Use: 18% | Physiological Mediators: 7% | Psychological Moderators: 5%

HOMA-IR = Homeostatic Model Assessment for Insulin Resistance | DERS = Difficulties in Emotion Regulation Scale | SCS-B = Self-Compassion Scale